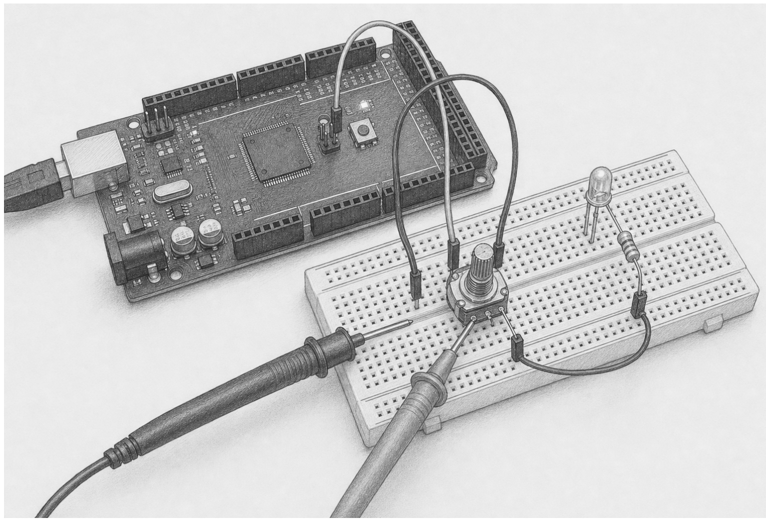
Make every transformation visible

Lesson 008: calibrate and stabilize a sampled signal

Predict → sample → map → average → hold → observe



Orientation plate. Use the pencil view to identify the three observable places: the knob, **TP-A0**, and the brightness LED. It is not a pin map. The exact net list and schematic below are authoritative.

Question	Can explicit, replayable transformations reduce visible jitter without hiding what the circuit measured?
Time	75–100 minutes
Prerequisites	Lesson 007; integer ranges; explicit status handling
You need	Mega 2560, USB cable, breadboard, 10 kΩ linear potentiometer, LED, resistor of at least 220 Ω, jumpers, multimeter
Evidence	TP-A0 voltage, raw and transformed worksheet, D6 duty/brightness, D13 acquisition witness, safe-state record
Safety class	E1: USB-powered Mega and passive low-current parts only
Status	Host-verified and experimental; physical acceptance remains open

Safety boundary

Remove USB before wiring or moving a probe connection. Use a 10 kΩ potentiometer between Mega 5 V and GND; connect only its wiper to A0. Drive the external LED from D6 through its own resistor of at least 220 Ω. All grounds are common.

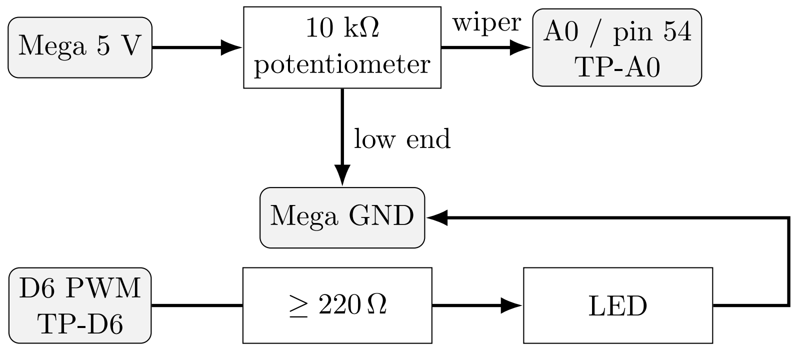
Filtering is interpretation, not electrical protection. It cannot make an over-voltage safe, prove a wire is sound, or turn a floating input into valid evidence. A rail endpoint can be a real knob position or a fault. A disconnected A0 can wander through plausible values.

Stop for heat, odor, smoke, USB disconnects, a voltage outside 0–5 V, an unidentified potentiometer terminal, or an unknown resistor. Disconnect USB upstream; do not move powered breadboard wiring.

Check — before wiring
☐ USB removed ☐ pot resistance and terminals identified ☐ LED resistor measured ☐ common ground traced

1. Reuse a controlled source

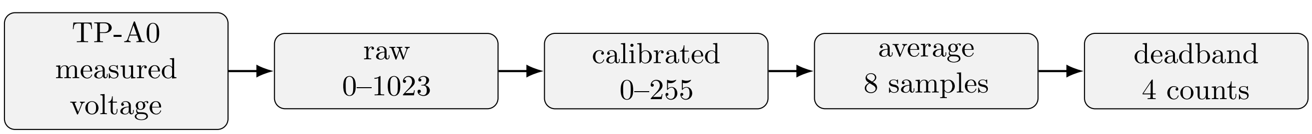
A controlled potentiometer is deliberately retained from Lesson 007. It lets you hold, step, and repeat the input before Lesson 009 adds a photoresistor's environmental variability.



From	Through	To	Purpose
Mega 5 V	pot high terminal	—	known source
pot wiper	TP-A0	Mega A0	sampled position
pot low terminal	—	Mega GND	divider return
Mega D6	resistor, LED anode	LED cathode to GND	brightness witness
Mega D13	built-in LED	—	acquisition witness

2. Predict the evidence chain

The circuit and program make five different claims. Do not collapse them into "the LED works."



D13 on supports the separate claim that all three hardware owners initialized. D6 brightness and its PWM waveform expose the final held value. TP-A0 remains available to challenge every software interpretation.

Predict. Before power, write what should happen after a quick knob step and what should happen while the knob is held still:

3. Calibrate observations, not ideal endpoints

The canonical example maps observed readings 80 through 940 onto duty 0 through 255 and clamps values beyond those observations. These numbers are a repeatable exercise, not universal facts about your potentiometer.

For an increasing calibration,

$$y = y_{\min} + \frac{(x - x_{\min})(y_{\max} - y_{\min})}{x_{\max} - x_{\min}},$$

with integer rounding. `LinearCalibration` validates the interval and returns `Result<uint16_t>` because an invalid configuration has no meaningful mapped result.

```
const adk::LinearCalibrationConfig calibrationConfig{
    80, 940, 0, 255, true};

adk::LinearCalibration calibration (calibrationConfig);

const adk::Result<uint16_t> mapped = calibration.map (raw);
if (!mapped.ok ())
{
    stopSafely ();
}
```

Position	TP-A0	Raw observation	Mapped prediction	Mapped result
low stop			0 after clamp	
quarter				
middle				
three-quarter				
high stop			255 after clamp	

Measure Mega 5 V and TP-A0 with power on only after placing probes while unpowered. The ideal relation is

$$\text{raw} \approx \frac{V_{\mathrm{TP-A0}}}{V_{\mathrm{ref}}} \times 1023.$$

Use measured V_{ref}, not an assumed perfect 5.000 V.

4. Advance state only from supplied samples

`MovingAverage` owns a fixed array of at most 32 readings. The example uses eight. It has no clock and takes no hidden sample: each `addSample()` advances it exactly once. Early results average the samples received so far.

`Deadband` publishes the first sample, then retains that value until a new sample differs by at least its width. It prevents small visible output changes; it does not erase raw evidence.

```
adk::MovingAverage average        (8);
adk::Deadband      brightnessHold (4);

const adk::Result<uint16_t> smoothed =
    average.addSample (mapped.value ());
const uint16_t stable =
    brightnessHold.addSample (smoothed.value ());
```

Order matters. Mapping first gives the deadband a meaning in output duty counts. Averaging before the deadband lets a real sustained change accumulate. Reversing these objects would describe a different instrument.

5. Replay a deterministic sample stream

Calculate the eight-sample, rounded moving average, then apply a deadband of four. The initial published value is 100.

Step	Supplied sample	Window contents	Rounded average	Held output
1	100	100	100	100
2	102	100, 102		
3	101	100, 102, 101		
4	104	100, 102, 101, 104		
5	112	100, 102, 101, 104, 112		
6	116	100, 102, 101, 104, 112, 116		
7	120	100, 102, 101, 104, 112, 116, 120		
8	124	all eight		

Replay the same values twice. The result must match at every step. Change only one sample and identify the first output that can change. This is the central testability property: the state is a function of configuration and supplied sample history, not processor speed or wall-clock timing.

6. Read the narrative run loop

The canonical sketch introduces owners before transformations. Setup validates the software configuration, acquires D13, A0, and D6, then lights D13. The loop reads as observe, decide, actuate:

```
potentiometer.update ();

adk::PwmOutput::Duty brightness = 0;
if (!chooseBrightness (brightness))
{
    stopSafely ();
    return;
}

if (!brightnessLed.write (brightness).ok ())
{
    stopSafely ();
}
```

The 20 ms sampling schedule belongs to the application. Calibration, averaging, and deadband remain useful in host tests without Arduino time. Shutdown turns off D6, makes A0 and D6 high impedance, releases all claims, and turns off D13. Destruction repeats that cleanup safely.

7. Prove the transforms on the host

Write deterministic tests before interpreting the breadboard:

1. invalid and reversed calibration intervals;
2. exact endpoints, midpoint rounding, clamping, and unclamped rejection;
3. moving-average windows 0, 1, 8, 32, and 33;
4. partial fill, full fill, replacement, reset, and 16-bit extremes;
5. deadband equality, just inside, just outside, rising, falling, reset;
6. two identical sample streams with identical result/status traces;
7. A0 claim conflict and D6 timer conflict before any unintended write;
8. repeated shutdown and destruction from an active circuit.

```
make host-test
make quality-fast
make arduino-Lesson008SampledSignal
```

Compilation and simulation do not establish physical acceptance.

8. Upload and observe without depending on Serial

Identify the board and port, compile the one canonical sketch, upload it, and optionally capture Serial only if a later experiment adds textual records:

```
make boards
make arduino-Lesson008SampledSignal
make upload EXAMPLE=Lesson008SampledSignal PORT=/dev/ttyACM0
make monitor PORT=/dev/ttyACM0 BAUD=115200
```

The required evidence is circuit-native:

1. D13 off at reset, then steadily on: resource-acquisition witness.
2. TP-A0 voltage follows the knob: primary input evidence.
3. D6 duty and LED brightness follow a sustained change: output evidence.
4. Small TP-A0 movement may leave D6 unchanged: deadband evidence.
5. A fast step reaches D6 gradually: moving-average evidence.
6. After the published shutdown procedure, D13 and the LED are off; a meter or logic analyzer, not darkness alone, checks safe state.

Experiment	Prediction	Observation	Interpretation
hold at middle	TP-A0 and D6 settle		
small nudge	TP-A0 moves; D6 may hold		
large step	TP-A0 jumps; D6 ramps		
return step	same direction-independent rules		
shutdown	D13/LED off; pins released		

9. Diagnose by stage

Observation	Next measurement	What it can establish
D13 stays off	inspect status path and unpowered wiring	acquisition failed; not which owner failed without recorded status
D13 on, TP-A0 fixed	measure wiper and both pot ends to GND	distinguishes source wiring from later transforms
TP-A0 moves, D6 fixed	inspect expected mapped/average/deadband worksheet; probe TP-D6	separates decision from LED wiring
D6 waveform changes, LED fixed	remove power; inspect polarity and resistor	the command reached the pin but the load path is suspect
rail reading	inspect all three pot terminals unpowered	may be endpoint or fault; filtering is not evidence either way
plausible wandering input	power off; check A0 continuity	a disconnected input can float plausibly

10. Extend the design

1. Reduce the average window from eight to two. Predict response time and visible jitter before running it.
2. Increase the deadband from four to sixteen. Which deliberate knob movements disappear?
3. Swap average and deadband in a host-only experiment. Explain why it is a different policy.
4. Record ten raw endpoint readings, choose observed calibration bounds, and justify the margin retained at each end.
5. Replace the potentiometer samples with a recorded photoresistor trace on the host. Do not change hardware yet. Which parameters need reconsideration for Lesson 009?

Bench acceptance record

Board and supply
Meter / analyzer
Measured 5 V reference
Measured pot and LED resistor
Observed calibration bounds
D13 acquisition observation
TP-A0 observations
TP-D6 / brightness observations
Shutdown pin evidence
Deviations, reviewer, date

Check — exit

☐ source circuit checked unpowered ☐ every transformation predicted ☐ deterministic replay passed ☐ non-Serial evidence recorded ☐ resource and safe-state evidence kept separate

Companion material. Use the HTML lesson for current API signatures, canonical source links, errata, and searchable procedures. Use this PDF for predictions, hand calculations, stage-by-stage diagnosis, and the physical acceptance record. Until that record is completed by a person at the named bench, this lesson remains host-verified.

Arduino Mega 2560 documentation Microchip ATmega2560 datasheet